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EXTENDED ABSTRACT

A Simple CRPS-based Ensemble Extension of Deterministic Precipitation Downscaling Models

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1. Introduction and Objectives

Deep Learning (DL) methods have shown skill in capturing complex non-linear predictor–predictand relationships for climate downscaling [Rampal(2024)]. However, deterministic and probabilistic models currently used for downscaling climate projections (e.g., MSE and Bernoulli-Gamma loss functions), exhibit important limitations in reproducing extremes and in properly quantifying uncertainty [González-Abad(2025)]. In this work, we propose a simple CRPS-based extension that significantly outperforms these traditional approaches.

2. Methods and Data

Our study follows the experimental framework of [González-Abad(2025)], but adopts a Vision Transformer (ViT) architecture [Dosovitskiy(2020)]. The large-scale predictors are partitioned into patches, embedded, and passed through a series of Transformer blocks. The resulting patch representations are then upscaled to match the resolution of the local-scale predictand.

To represent deterministic approaches, we train the ViT using the MSE and asymmetric (ASYM) loss functions. While MSE is the most standard choice, the ASYM loss, identified in [González-Abad(2025)] as the best-performing deterministic loss, weights observations based on a Gamma distribution fitted to the training data, thereby giving greater importance to extreme values. As a representation of standard probabilistic approaches, we include the Bernoulli–Gamma method [Baño-Medina(2022)], which predicts the parameters of a Bernoulli–Gamma distribution and minimizes the negative log-likelihood. As a contribution of this work, we evaluate the CRPS and CRPS-Spectral loss functions [Nordhagen(2025)]. These losses extend the mean absolute error to probabilistic forecasting, with the spectral variant incorporating frequency-domain information to reduce spatial inconsistencies. For the CRPS-based models, noise is injected through conditional layer normalization.

3. Preliminary Results and Discussion

Figure 1 presents several test-set evaluations addressing key aspects of this study. The top row shows the Power Spectral Density (PSD), averaged over all test days, a histogram highlighting extremes using a logarithmic y-axis, and the spread–skill plot for the ensemble approaches. The bottom row shows the

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downscaled precipitation fields of a particular day of the test set (2017-10-18) for the ground truth and the different models.

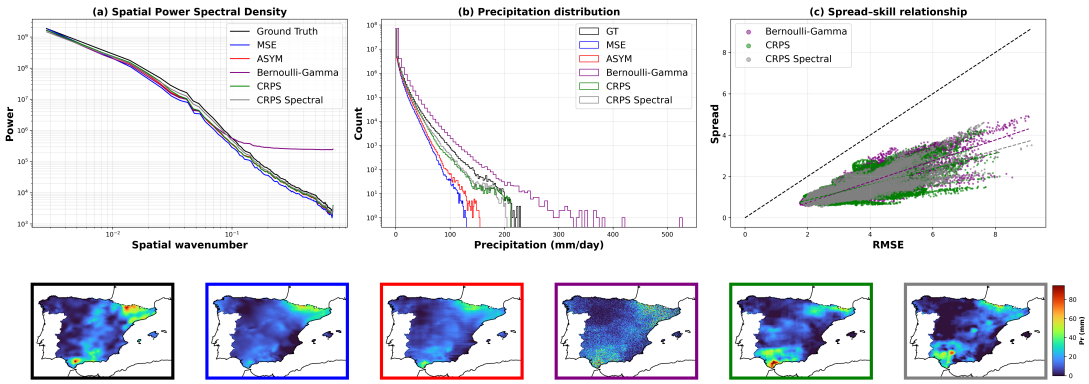

Figure 1. *A comparison of observed and downscaled precipitation (test period). The top row shows the power spectral density, histogram (log y-axis) and spread–skill. The bottom row shows predictions for a particular day from each model (colored borders) alongside the observations (black).*

Regarding the PSD, most models reproduce the frequency content well across scales, with CRPS and CRPS-Spectral performing best. Bernoulli–Gamma overestimates high frequencies, likely due to the independence of its modeled distributions, which also leads to spatial inconsistencies in the daily fields. ViT models trained with CRPS-based losses capture extreme events missed by deterministic approaches—and overestimated by the Bernoulli–Gamma model while maintaining spatial consistency. This suggests that CRPS-based training effectively balances the representation of extremes with realistic spatial structure. Most models exhibit similar calibration behavior in the spread–skill plots, with Bernoulli–Gamma performing slightly better among them. Regarding the spatial fields, CRPS-Spectral most realistically reproduces precipitation structures over the Iberian Peninsula.

4. Future Work

We are currently expanding the intercomparison to include state-of-the-art methods, such as conditional GANs and diffusion-based models. We are also conducting a more thorough evaluation of model calibration, an aspect we believe has been overlooked in the literature. Finally, future work will focus on applying these methods to GCM downscaling, to assess their ability to extrapolate to previously unseen future climate conditions.

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